

Román

* Escobedo

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I am a physicist with a strong foundation in mathematics. My interests lie in applications of abstract mathematics in the physical world. So, I am looking to broaden my experience and knowledge in different areas of mathematics, e.g., topology and geometry. My inclination is towards the mathematical foundations of Quantum Field Theories. However, I am also passionate about more practical topics like Topological and Geometrical Data Analysis applied in biology, e.g., neuroscience.

education

MSc. in Mathematics Technical University of Munich	<i>2024-current</i>
BSc. in Physics and Mathematics Instituto Politecnico Nacional	<i>2018-2023</i>
Vocational Diploma in Graphic Design Instituto Politecnico Nacional	<i>2015-2018</i>

research experience

Research Assistant BEIFI | Instituto Politecnico Nacional *march2023-aug2023*

Assistant in the project "Symmetries of non-linear systems and quantization":

- Writing and designing didactic notes on complex topics in mathematics.
- Gathering experimental data and using SciPy Python's Library to model and predict the physical behavior observed.

Research Assistant BEIFI | Instituto Politecnico Nacional *jun2022-nov2022*

Assistant in the project "Solitons and Symmetries: Korteweg-De Vries equation's symmetry group":

- Scientific reports writing.
- Elaboration of support graphics of complex mathematical and physical concepts.

industrial experience

Quality & Production Internship | Industrias de Kores de México *nov2022-feb2023*

I worked with the quality and production team to analyze the data they've been gathering along the last years:

- Statistical analysis of the process with Excel and statistical software Minitab.
- Updated specifications of the products, resulting in an improvement of 60% in the production chain.
- Cleaned and maintained the database, also updating the report sheets with the most impactful data.

languages

Spanish (mother tongue) **English** (TOEFL iBT 98/120) **German** (beginning to learn)

projects

Hopf Fibration and the Qubit | Undergraduate Dissertation Project *sept2023-current*

I'm developing a visualization of the Bloch sphere, the pure state space of a quantum two-level system, which normally will be hard to visualize. This is being done by applying the Hopf fibration to reduce its dimensions and discovering topological information on the phase factors of the states.

Connections between Differential Geometry of Curves and Classical Mechanics | Disclosure Paper

jan2023-oct2023

I developed, with Dr. Pablo Paniagua and Dr. Francisco Turrubiates, a disclosure paper that shows and exemplifies connections between physical quantities and geometrical quantities of curves. For example, we found expressions for the curvature, torsion, and the Frenet-Serret orthogonal frame solely in terms of the force, and momentum of the system. The final work was presented at Reunión Nacional Académica de Física y Matemáticas.

certifications

Google Advanced Data Analytics | Google Professional Certificates
Analyzing and Visualizing Data in Looker | Google Cloud Training
Intro to Deep Learning | Kaggle
Machine Learning | Centro de Investigacion en Computación - IPN

sept2023-current
sept2023
aug2023
july2023

technical skills

software	Microft Office	Illustrator	programming	Python(numpy, pandas, scipy)
	Minitab	Photoshop		C/C++
	Tableau	InDesign		SQL/postgreSQL
	PowerBI	Matlab		HTML/CSS

awards

3rd Place - Company of the year LATAM | JA LATAM

2017

1st Place - Company of the year Mexico | JA Mexico

2017

International competition of entrepreneurship held in Brazil, where my team and I represented México with our project and company.